



**UNIVERSITY OF BENIN, BENIN CITY EDO STATE**

**IMPLEMENTATION AND SIMULATION A TEMPERATURE  
MONITORING CIRCUIT USING PIC18F452 IN PROTEUS**

**DONE BY**

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## Introduction

Temperature sensing is a fundamental aspect of many electronic systems, from climate control to industrial automation. Simulating these sensors before physical implementation allows engineers to validate designs, troubleshoot issues, and optimize performance. Proteus, a powerful electronic design automation (EDA) tool, provides a virtual environment to model and test circuits involving temperature sensors like the LM35 or thermistors.

This report presents the simulation of a temperature sensing system using the LM35 temperature sensor in Proteus software. The objective is to monitor temperature variations and display the result using a 16x2 LCD driven by a microcontroller (PIC18F452). This kind of setup is useful in environmental monitoring and industrial control systems.

The simulation of temperature sensors in Proteus provides a virtual platform to design and test temperature measurement systems without the need for physical hardware. This enables faster development and troubleshooting in embedded applications, especially where microcontrollers interface with analog sensors.

## Background of Study

Temperature sensing is vital in many applications ranging from weather stations to embedded control systems. The LM35 is a popular analog temperature sensor that outputs  $10 \text{ mV}/^{\circ}\text{C}$ . When interfaced with a microcontroller that has an ADC (Analog-to-Digital Converter), the sensor readings can be converted and displayed in real-time.

Temperature sensors, like thermistors or LM35, output electrical signals proportional to temperature. To accurately read these signals, microcontrollers use ADCs (Analog-to-Digital Converters). This report contains simulation of both sensors and microcontroller peripherals together, enabling validation of circuits under different temperature conditions.

- ◆ Temperature sensors convert thermal energy into electrical signals. Common types include:
- ◆ Thermistors: Resistance changes with temperature.

◆ LM35: Provides a linear voltage output proportional to temperature in Celsius.

This study focuses on simulating an LM35 sensor circuit with a microcontroller (PIC16F877A) to read and display temperature values.

## Methodology

The simulation was conducted using the following steps:

### Tools & Components

- Proteus Design Suite
- LM35 temperature sensor
- PIC18F452 microcontroller
- 16x2 LCD Display
- Resistors
- Capacitors
- Potentiometer (for LCD contrast)
- Crystal
- 5V DC Power source

## Circuit Design

1. Create a new project in Proteus.
2. Place components: LM35, microcontroller, LCD, and supporting elements.
3. Connect the LM35 to an analog input pin of the microcontroller.
4. The LCD was connected to PORTD (data lines) and PORTE (control lines).
5. The microcontroller was programmed using mikroC and the .hex file was loaded into Proteus.

Here is the code used;

```
// Define LCD connections to PORTD
// This must match your Proteus circuit
sbit LCD_RS at RD1_bit;
sbit LCD_EN at RD2_bit;
sbit LCD_D4 at RD4_bit;
sbit LCD_D5 at RD5_bit;
sbit LCD_D6 at RD6_bit;
```

```

sbit LCD_D7 at RD7_bit;

// Define pin directions for the LCD connections
sbit LCD_RS_Direction at TRISD1_bit;
sbit LCD_EN_Direction at TRISD2_bit;
sbit LCD_D4_Direction at TRISD4_bit;
sbit LCD_D5_Direction at TRISD5_bit;
sbit LCD_D6_Direction at TRISD6_bit;
sbit LCD_D7_Direction at TRISD7_bit;

// --- Main Program ---
void main() {
    unsigned int adc_value;
    float temperature;
    char temp_text[16]; // Buffer to hold the temperature string

    // Configure AN0/RA0 as an analog input
    // The rest of PORTA will be digital
    ADCON1 = 0x0E;
    TRISA = 0x01; // Set RA0 as an input pin

    // Initialize the LCD library
    Lcd_Init();
    Lcd_Cmd(_LCD_CLEAR); // Clear the display
    Lcd_Cmd(_LCD_CURSOR_OFF); // Turn the cursor off

    Lcd_Out(1, 1, "Temperature:"); // Write "Temperature:" on the first line

    while(1) {
        // Read the analog value from channel 0 (AN0)
        adc_value = ADC_Read(0);

        // Convert the 10-bit ADC value to temperature in Celsius
        // Formula: Temperature = (ADC_Value * Vref / 1024) / (10mV/°C)
        // Simplified: Temperature = (adc_value * 500.0) / 1023.0
        temperature = (adc_value * 500.0) / 1023.0;

        // Convert the floating-point temperature value to a string
        FloatToStr(temperature, temp_text);
        temp_text[4] = 0; // Keep only one decimal place (e.g., "25.3")

        // Display the temperature on the second line of the LCD
        Lcd_Out(2, 1, temp_text);
        Lcd_Chrcp(' '); // Write a space
        Lcd_Chrcp(223); // Write the degree symbol '°'
        Lcd_Chrcp('C'); // Write 'C'

        Delay_ms(500); // Wait for half a second before the next reading
    }
}

```

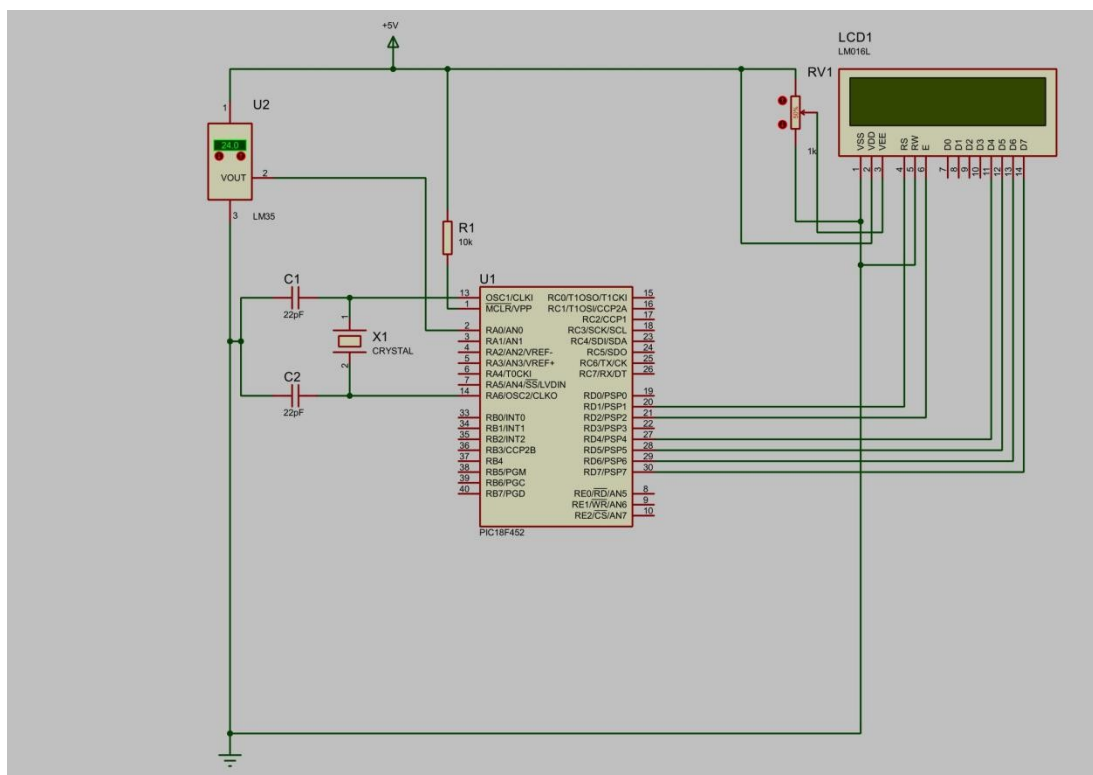
6. The program converted the ADC value into temperature using the formula:

$$\text{Temperature} = (\text{ADC\_value} \times 5 \times 100) / 1023$$

7. The value was displayed on the LCD.

## Simulation

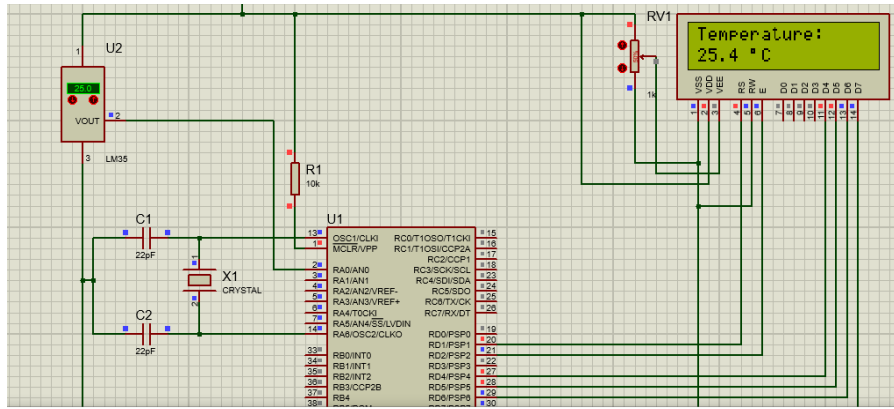
- ✓ Start the simulation in Proteus.
- ✓ Adjust the temperature input manually or via virtual controls.
- ✓ Observe the voltage output from LM35 and the corresponding temperature displayed on the LCD.



**The Proteus Design Of Temperature Sensor**

## Result

The simulation successfully showed real-time temperature readings on the LCD screen. The ADC conversion, signal processing, and display routines functioned correctly, verifying the design's accuracy and responsiveness under temperature changes.



Upon running the simulation:

- ❖ The LCD displayed real-time temperature values based on the voltage output from the LM35.
- ❖ For example, an LM35 output of 0.25V displayed 25°C.
- ❖ Varying the input caused the displayed temperature to update accordingly.

It also demonstrated;

- Accurate voltage output from LM35 corresponding to temperature changes.
- Real-time temperature display on the LCD.
- Microcontroller correctly interpreting analog signals and converting them to Celsius values.

For example, at 30°C, the LM35 output was approximately 0.30V, and the LCD displayed “Temp: 30°C”.

## Conclusion

The simulation successfully demonstrated the functionality of a temperature sensing system using the LM35 sensor and PIC microcontroller. This design can be extended to include data logging, alarms, or wireless transmission in practical applications.